Sprint 2 Review
Presentation

Group [1]

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ZERO_ HUNGER LAB

TILBURG UNIVERSITY

Objectives of Current Sprint

- Explore different impact areas to find insightful directions
- Narrow the individual searches and share with the group
- Compare different ROs and choose one to pursue deeper

- Understand and improve the INFLOW-AI model for flood prediction

Sprint Summary (from sprint 2 onward)

- Our main goal was to find a RO worth pursuing, balancing feasibility, novelty and stakeholder-alignment
- We explored several areas of impact and after deepening into them we managed to define a unique RO
- Data was very limited -> we had to drop many areas and ROs that seemed logical / interesting
- Be more consistent with internal deadlines, initiative taking.

Methodological Angle (Poster/Report):
Problem definition & research objective

- Anticipatory action for floods needs to know how much farmland is under water: ZOA's example trigger is a flood “inundating cropland”.
- In South Sudan that number comes from overlaying public flood maps on public cropland maps.
- Nobody has checked how accurate those maps are where ZOA works: along the Lol river (Aweil) and in Bor South.

Research objective

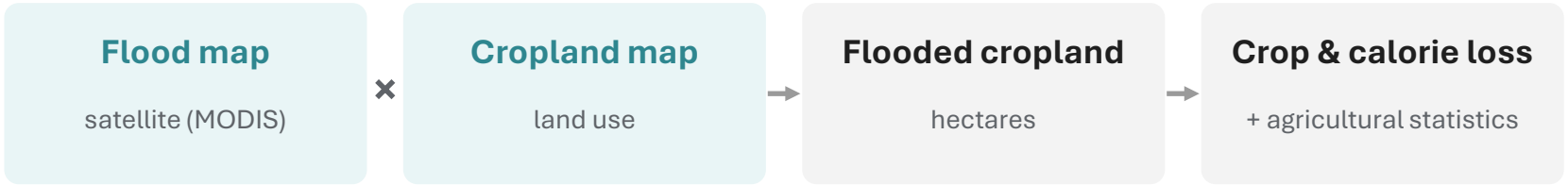
Evaluate how accurately public satellite data measure cropland, and cropland flooding, in ZOA's areas, and which conditions go with flooding of farmland.

Five Aweil counties + Bor South

Reference season 2022, flood record 2017–2025

≈450 blind, double-labelled 30 m points

Methodological Angle (Poster/Report):
The method we build on



Our RO tests these two inputs where ZOA works

The authors' own warning

"Remote sensing data by itself lacks the level of accuracy necessary in a comprehensive food security assessment."

Changing one input by ±20% changed the estimated losses by 12%.

Same overlay, used today

WFP ADAM reports flooded cropland per county (flood map × NASA GFSAD30) with no accuracy statement.

Kerner et al. (2024) checked 11 cropland maps in 8 African countries, not South Sudan.

Pacetti, Caporali & Rulli (2017), Advances in Water Resources 110:494–504 (Bangladesh 2007, Pakistan 2010). Kerner et al. (2024), Scientific Data, doi:10.1038/s41597-024-03306-z.

Methodological Angle (Poster/Report):
Why the problem is neglected

1 No ground truth

Accuracy studies skip South Sudan (Kerner et al. 2024: 8 countries, none of them South Sudan). We found no accuracy figure for flooded cropland there.

2 Clouds in the flood season

Optical flood maps are blind in July–September, exactly when fields flood. Radar sees through clouds but is not used in the standard overlay.

3 Small, shifting fields

Smallholder plots are small, move between years and lie fallow, so 10–30 m global maps easily miss them.

4 Numbers without error bars

Tools publish hectares without uncertainty; CFSAM's harvested area is computed from population and household statistics, not measured.

Methodological Angle (Poster/Report):
What unreliable data costs

35×

less flooded cropland in the standard overlay than CFSAM reports

South Sudan, Jun–Nov 2022: flood masks × ASAP ≈ 3,700 ha vs 130,000 ha damaged

0 ha

flooded cropland ever detected in Bor South

flood masks × ASAP, every season checked, in an area ZOA names as flood-prone (Q&A 2, slide 3)

70 / 77

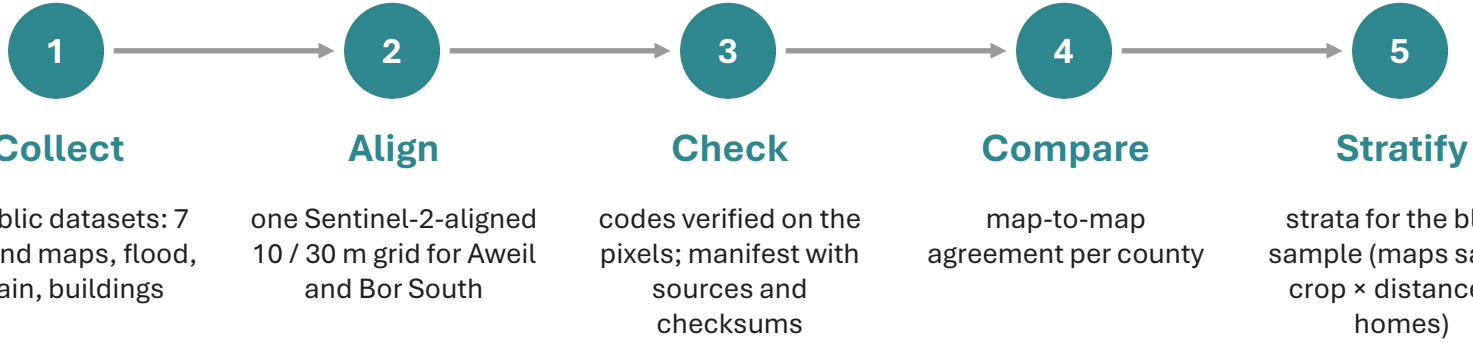
counties where the ASAP cropland map is below CFSAM's harvested area

national: ≈407,000 ha mapped vs 1.03–1.18 million ha harvested

What that means for anticipatory action

- A trigger on “flood inundates cropland” would rarely fire, even in seasons when harvests are lost.
- Aid follows the data, not the damage; and figures without uncertainty look more precise than they are.

Methodological Angle (Poster/Report): Progress and Results

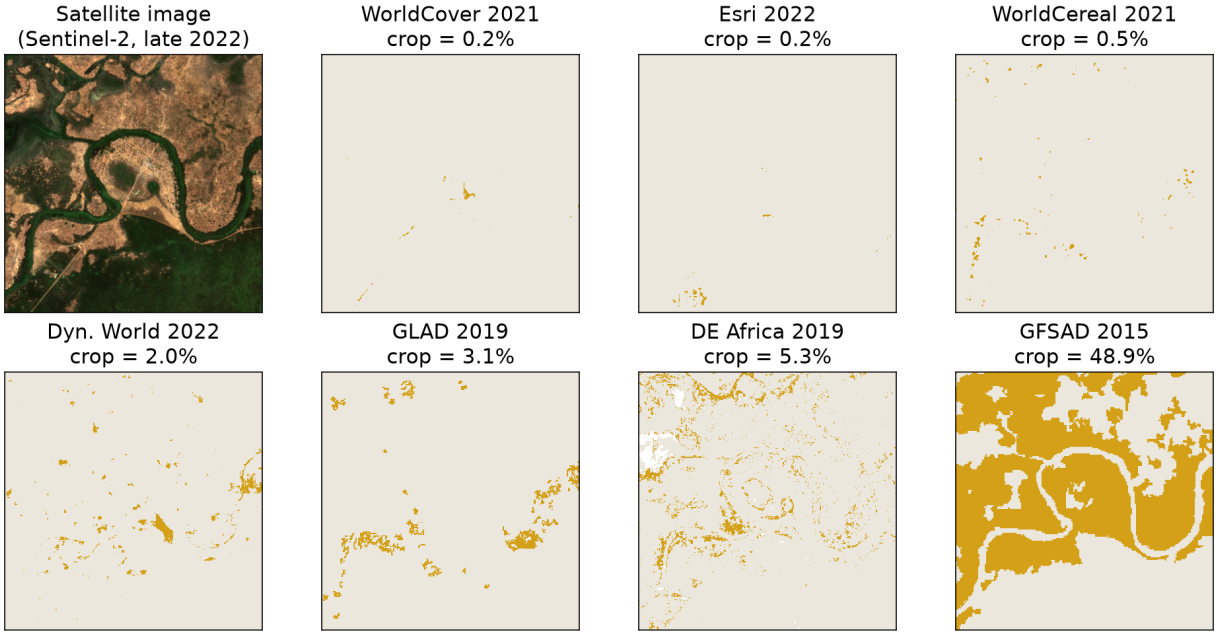


Done this sprint: steps 1–4 for both areas, scripted and re-runnable.

Next: Sentinel-1 radar flood layer for 2022, fix the strata, draw the sample, 100-point labelling pilot.

Methodological Angle (Poster/Report):

First results: seven maps, seven answers



0.2%

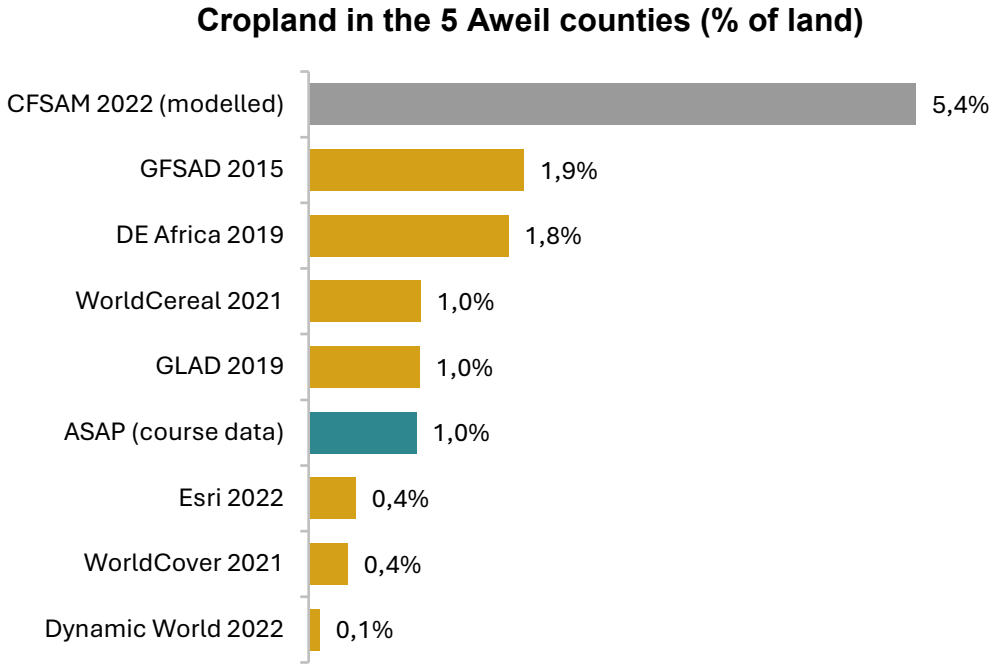
to

48.9%

of the same window called crop, depending on which map you ask

Same 6 × 6 km near Turalei (Twic county), outside our study counties so the labelling stays blind. Gold = map says crop.

Methodological Angle (Poster/Report):
First results: cropland per map in Aweil



1 ha

where all four main maps (2021–22) agree on crop, in 3 million ha

≤ 25%

best overlap of the crop areas of any two maps

5×

CFSAM's modelled harvested area vs the typical map

Crop share from 10 m pixels, counties' own area. CFSAM = 2022 main-harvest cereal area, a modelled estimate (context, not a reference). Agreement is between maps, not accuracy.

Methodological Angle (Poster/Report): Mistakes and what you have learned

- Data quality matters more tan what we thought
- Do not assume relationships that might seem logical. Always prove claims with data (datasets OR research evidence)

Stakeholder Angle (Meetings): Progress and Results

Harvest failure is the priority impact

slide 6

and ≈80% of rural communities live mainly from agriculture (slide 33)

“The dataset might not capture agricultural land”

slide 33

ZOA already suspects the undercount our numbers show

Triggers like a flood “inundating cropland”

slides 31, 40

only usable if the cropland and flood data are reliable

Show the reliability, not a perfect forecast

slides 9–11, 32

a rigorous small sub-problem beats a superficial full one (slides 5, 29)

Stakeholder Q&A 2 (ZOA / Zero Hunger Lab, 21 Sep 2026). Our RO covers ZOA's two areas: the Lol river in Aweil and Bor South (slide 3).

Stakeholder Angle (Meetings):
Mistakes and what you have learned

- Not align our objective with the stakeholder needs
- Understanding the problem counts more than assuming things and give a naïve forecast

Societal-Legal-Ethical Angle (Essay):
Progress: the problem as an SLE question

The problem

- ZOA's example trigger: a flood “inundating cropland” (Q&A 2, slides 31, 40).
- That number comes from a public flood map laid over a public cropland map.
- Nobody has checked those maps along the Lol river (Aweil) or in Bor South.

Why it matters

- ≈80% of rural communities live mainly from farming, mostly smallholders (slide 33).
- Harvest failure is ZOA's priority flood impact (slide 6).
- ZOA itself warns: “the dataset might not capture agricultural land” (slide 33).

SLE question (draft, for the team)

Whose farmland counts? How the choice of a satellite map decides which flood-hit farmers anticipatory action can see.

Lenses from the lectures

“Technical choices = SLE choices” (L1 p. 5)

“Whose way of ‘seeing’ counts?” (Jasanoff, L1 p. 7)

“Which communities? Who? Where?” (Sen, L2 p. 11)

Stakeholder Q&A 2 (ZOA / Zero Hunger Lab, 21 Sep 2026). L1 = SLE lecture 1 (Petročnik), L2 / L3 = SLE lectures 2 and 3 (van Maanen).

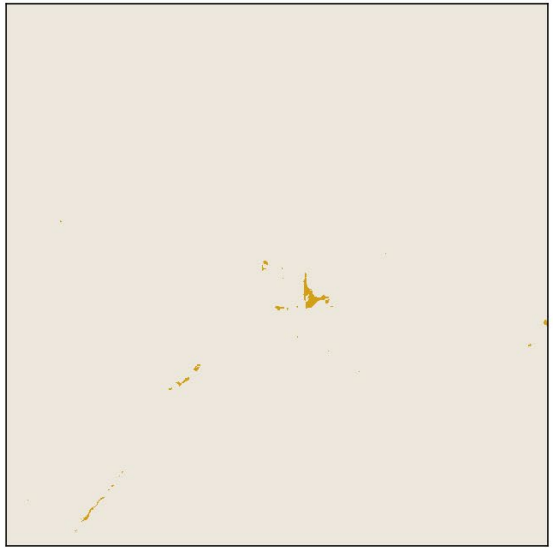
Societal-Legal-Ethical Angle (Essay):

Result: the map decides who is seen

Satellite image (late 2022)



WorldCover 2021: 0.2% crop



GFSAD 2015: 48.9% crop



Same fields, near Turalei (outside our study area): one map sees almost no farms, the other sees half the land.

Aweil counties: 1 ha of cropland if all four main maps must agree, ≈47,000 ha if one is enough, ≈140,000 ha if any of seven counts.

Gold = map says crop. Reading: data is socially constructed, not simply “seen” (L1 p. 7); unrepresentative data create blind spots (L1 p. 10). Agreement between maps is not accuracy.

Societal-Legal-Ethical Angle (Essay): Prior paper and why it is neglected

The paper we build on

Pacetti, Caporali & Rulli (2017): flood map × land-use map → lost crops and calories (Bangladesh 2007, Pakistan 2010).

Their own warning: "Remote sensing data by itself lacks the level of accuracy necessary in a comprehensive food security assessment."

WFP ADAM uses the same overlay for South Sudan, with no accuracy statement.

Why the problem is neglected

- No ground truth: accuracy studies skip South Sudan (Kerner et al. 2024: 8 countries, not this one).
- Clouds hide the fields in July–September, exactly when they flood.
- Smallholder plots are small, move and lie fallow: global maps miss them.
- Hectares are published without error bars, so the gap stays invisible.

Leads for the essay: Who writes what, and for what reason? (L3 p. 18). Do exact-looking hectares serve reporting to donors more than decisions in the field? (logic of audit, L2 p. 4)

Pacetti et al. (2017), Advances in Water Resources 110:494–504, quote p. 501. Kerner et al. (2024), Scientific Data, doi:10.1038/s41597-024-03306-z.

Societal-Legal-Ethical Angle (Essay):
What unreliable data costs people

1 Farmers left out

A field missing from the map has no flooding and no household to target. Bor South: 0 ha flooded cropland ever detected, yet its roads close 77–92% of weeks in Sep–Dec.

Targeting (Chavez-Gonzalez et al., L3 p. 16); blind spots (L1 p. 10)

2 Triggers that stay silent

2022: the standard overlay finds ≈3,700 ha of flooded cropland in South Sudan; CFSAM reports 130,000 ha damaged. The trigger sees ≈3%.

Timing and activity: acting too late or not at all (L3 p. 16)

3 False certainty

Hectares without error bars look exact. ZOA asked for the risks of the data so they can choose between misses and false alarms (Q&A 2, slides 2, 9–11).

Data is not self-explanatory (Letz & Maxwell, L3 p. 17)

4 Hectares are not hunger

Flooded cropland is one factor. The same flood hits IDPs, hosts and pastoralists differently (Q&A 2, slide 38).

Do not use one factor; entitlements (Sen, L2 pp. 8–10)

Societal-Legal-Ethical Angle (Essay): Mistakes and what we have learned

Mistake	What we learned
We started from what a model can predict (a flood forecast)	ZOA decides about crops and people; tools can change the problem (L3 p. 11). We check the data first.
We took one cropland map as the truth	Seven maps give seven answers: a map is a way of seeing (L1 p. 7)

Lesson: report every number with its uncertainty and its blind spots, and write down our assumptions (Chavez-Gonzalez et al., L3 p. 16).

Group Peer Feedback

As a group, use the corresponding feedback from Feedback Fruits (in Canvas) to select one positive outcome, and one action or task to improve. An improvement can also touch on any aspect of the course that you want to approach more efficiently.

Positive

Add one group positive feedback here

Improvement

Add one, or a summary of different approaches that you will implement to i.e., meetings/teacher feedback/stakeholder discussions...